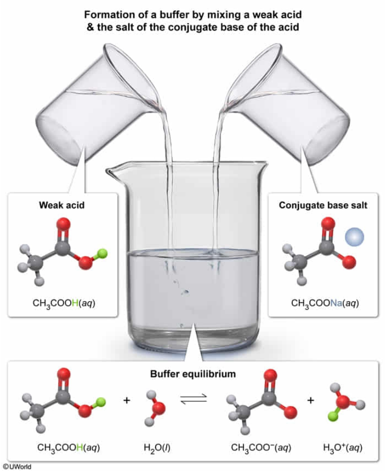
Which of the following compound pairs, dissolved into solution at equal concentrations, will function as a buffer?

- A. HNO_3(*aq*) and $NaNO_3$(*aq*) [16%]
- ✔ B. CH_3COOH(*aq*) and CH_3COONa(*aq*) [71%]
- C. NaBr(*aq*) and NaCN(*aq*) [2%]
- D. NaOH(*aq*) and NaCl(*aq*) [8%]

Correct
71% answered correctly

Explanation:



Buffers are solutions that resist changes in pH when small amounts of acid or base are added. Buffers consist of a **mixture** of either a weak acid and a salt of its conjugate base, or a weak base and a salt of its **conjugate acid**. The acidic component of a buffer mixture can neutralize any added base, and the basic component of a buffer mixture can neutralize any added acid. As a result, large changes in pH are impeded.

For example, acetic acid (CH_3COOH) is a weak acid, and its conjugate base is the acetate anion (CH_3COO^-). As such, a buffer could be made by mixing equal concentrations of each into solution. A salt such as sodium acetate (CH_3COONa) could function as the source of acetate ions. Once formed, the buffer would establish the following equilibrium:

CH $_3$ COOH +

Any strong base (OH^-) added to the solution would be neutralized by the CH_3COOH to form more acetate ions:

C H $_3$ COOH +

Similarly, any strong acid (H_3O^+) added to the solution would be neutralized by the CH_3COO^- to form more CH_3COOH:

C H $_3$ COO $^-$

Therefore, an aqueous mixture of CH_3COOH and CH_3COONa will function as a buffer.

(Choice A) $NaNO_3$ is the salt of the conjugate base of HNO_3, but HNO_3 is a strong acid and strong acids are not suitable for making buffers.

(Choice C) NaBr with NaCN is a mixture of two salts. A buffer requires a weak acid (or a weak base) to be present with the salt of its corresponding conjugate base (or conjugate acid).

(Choice D) NaOH is a strong base, and strong bases are not suitable for making buffers. Moreover, NaCl is not the salt of the conjugate acid of NaOH.

Educational objective:
Buffers are solutions that resist changes in pH when small amounts of acid or base are added. Buffers consist of a mixture of either a weak acid and a salt of its conjugate base, or a weak base and a salt of its conjugate acid.

Time spent:0 sec
Subject:
General Chemistry

QID:402372
System:
Solutions and Electrochemistry